



Year 12 Mathematics Extension 2 (4U)
HSC ASSESSMENT TASK 1
TERM 4, Week 6, 2006

Name: _____

Teacher: _____

Set By: DS

Tuesday 21 November 2006

- Attempt **ALL** questions.
- Marks may be deducted for careless, insufficient, or illegible work.
- Only Board approved calculators (**excluding** graphic calculators) may be used.
- Total possible mark is **40**.
- Begin each question on a new sheet of paper.
- **TIME ALLOWED: 60 minutes**

Complex Numbers and Mathematical Induction

Question 1 (12 marks) (Start a new page)

- (a) If $z = 3 - 2i$, and $w = -4 + i$, evaluate in the form of $x + iy$ the following:

(i) $2z - iw$ (ii) w^2 (iii) $\frac{w}{z}$ 1,1,1

(b) Simplify $\frac{1}{i} - \frac{3i}{1-i}$, expressing your answer in the form $a+ib$. 3

(c) (i) Write down the moduli and arguments of $-\sqrt{3} + i$ and $4 + 4i$. 2

(ii) Hence express in modulus/argument form $\frac{-\sqrt{3} + i}{4 + 4i}$ where $-\pi \leq \arg z \leq \pi$. 2

(iii) Evaluate $(-\sqrt{3} + i)^8$, expressing your answer in the form $a+ib$. 2

Question 2 (9 marks) (Start a new page)

(a) Solve the following for x and y : 3

$$\left(\frac{1+i}{1-i}\right)^2 + \frac{1}{x+iy} = 1+i.$$

(b) (i) Find the complex square roots of $-8 + 6i$. 3

(ii) Hence or otherwise solve $2z^2 + (1-i)z + (1-i) = 0$. 3

Question 3 (10 marks) (*Start a new page*)

- (a) Sketch on the Argand diagram the region which satisfies these expressions, **4**

$$2 \leq |z| \leq 3 \quad \text{and} \quad \frac{\pi}{4} < \arg(z - i) < \frac{3\pi}{4}.$$

- (b) Sketch on the Argand diagram and algebraically describe the locus of the point P representing z , given that $|z|^2 = z + \bar{z} + 1$. **3**

- (c) $z = x + iy$ is such that $\frac{z - i}{z + 1}$ is purely imaginary. Find the equation of the locus of the point P representing z . **3**

Question 4 (9 marks) (*Start a new page*)

- (a) Prove by the method of Mathematical Induction that $5^n \geq 1 + 4n$ for $n \geq 1$. **5**

- (b) Prove by the method of Mathematical Induction that $\arg z^n = n \arg z$ for $n \geq 1$. **4**

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$$1(a) \quad z = 3 - 2i \quad w = -4 + i$$

$$(i) \quad 2z - iw = 2(3 - 2i) - i(-4 + i) \\ = 6 - 4i + 4i - i^2 \\ = 6 + 1 = 7 \quad (1)$$

$$(ii) \quad w^2 = (-4 + i)^2 \\ = 16 - 8i + i^2 \\ = 15 - 8i \quad (1)$$

$$iii) \quad \frac{w}{z} = \frac{-4 + i}{3 - 2i} = \frac{(-4 + i)(3 + 2i)}{13} \\ = \frac{-14 - 5i}{13} \quad (1)$$

$$1(b) \quad \frac{1}{z} - \frac{3i}{1-i} = \frac{z}{-1} - \frac{3i(1+i)}{2} \\ = \frac{-2i - 3i - 3i^2}{2} \\ = \frac{-5i + 3}{2} \\ = \frac{3}{2} - \frac{5i}{2} \quad (3)$$

$$1(c) (i) \quad -\sqrt{3} + i = 2 \cos \frac{5\pi}{6} \quad (1) \\ 4 + 4i = 4\sqrt{2} \cos \frac{\pi}{4} \quad (1)$$

$$(ii) \quad \frac{-\sqrt{3} + i}{4 + 4i} = \frac{2 \cos \frac{5\pi}{6}}{4\sqrt{2} \cos \frac{\pi}{4}} \\ = \frac{1}{2\sqrt{2}} \cos \left(\frac{5\pi}{6} - \frac{\pi}{4} \right) \\ = \frac{\sqrt{2}}{4} \cos \frac{7\pi}{12} \quad (2)$$

$$1(c) (iii) \quad (-\sqrt{3} + i)^8 = 2^8 \left(\cos \frac{5\pi}{6} \right)^8 \\ = 256 \cos \frac{40\pi}{6} \\ = 256 \cos \frac{2\pi}{3} \\ = 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \\ = 256 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ = -128 + 128\sqrt{3}i \quad (2)$$

$$2(a) \quad \left(\frac{1+i}{1-i} \right)^2 + \frac{1}{x+iy} = 1+i$$

$$\frac{1+i-1}{1-i-1} + \frac{1}{x+iy} = 1+i$$

$$-\frac{2i}{-2i} + \frac{1}{x+iy} = 1+i$$

$$-1 + \frac{1}{x+iy} = 1+i$$

$$\frac{1}{x+iy} = 2+i$$

$$\frac{1}{x+iy} = \frac{2+i}{1}$$

$$x+iy = \frac{1}{2+i} = \frac{2-i}{5}$$

$$\text{Hence } x = \frac{2}{5}, y = -\frac{1}{5} \quad (3)$$

6(i). Let $-8+6i = (a+ib)^2$

Hence $a^2 - b^2 = -8$, $2ab = 6$

Hence $a^2 - \frac{9}{a^2} = -8$ $ab = 3$,
 $b = \frac{3}{a}$

$a^4 + 8a^2 - 9 = 0$

$(a^2 + 9)(a^2 - 1) = 0$

Hence $a = \pm 1$ Hence $b = \pm 3$ (3)

Hence $\sqrt{-8+6i} = 1+3i, -1-3i$

(ii) $2z^2 + (1-i)z + (1-i) = 0$

$z = \frac{-(1-i) \pm \sqrt{(1-i)^2 - 4(2)(1-i)}}{4}$

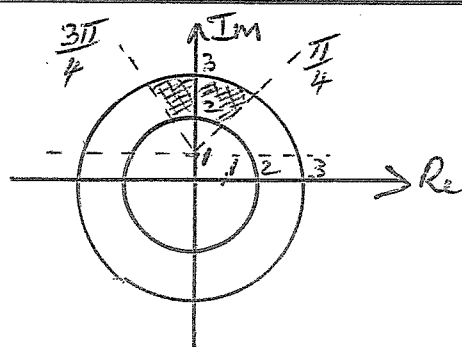
$= \frac{-(1-i) \pm \sqrt{-8+6i}}{4}$

$z_1 = \frac{-1+i + (1+3i)}{4} = i$

AND

$z_2 = \frac{-1+i + (-1-3i)}{4} = -\frac{1}{2} - \frac{i}{2}$ (3)

3(a)



$2 \leq |z| \leq 3$ and

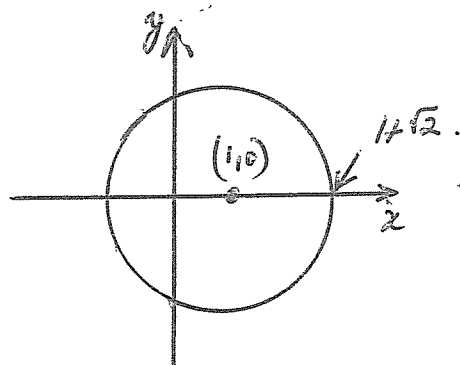
$\frac{\pi}{4} < \arg(z-i) < \frac{3\pi}{4}$

3(b) $|z|^2 = z + \bar{z} + 1$ Let $z = x+iy$

$x^2 + y^2 = 2x + 1$

$x^2 - 2x + y^2 = 1$

$(x-1)^2 + y^2 = 2$ which is a circle
 centre $(1,0)$ radius $\sqrt{2}$.



3(c) $\frac{z-i}{z+1} = \frac{x+i(y-1)}{(x+1)+iy} \times \frac{x+1-iy}{x+1-iy}$
 $= \frac{x(x+1) + y(y-1) + i(y-1)(x+1) - ixy}{(x+1)^2 + y^2}$

Now $\text{Re}\left(\frac{z-i}{z+1}\right) = 0$ Since $\frac{z-i}{z+1}$ is imaginary

Hence $x(x+1) + y(y-1) = 0$

$x^2 + x + y^2 - y = 0$

$\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}$

which represents a circle centre $\left(-\frac{1}{2}, \frac{1}{2}\right)$
 with radius $= \frac{1}{\sqrt{2}}$.

(3)

(4)

4(a) Prove by induction $5^n \geq 1+4n$
 $n \geq 1$
 $n \in \mathbb{J}^+$

step 1 for $n=1$

$$LHS = 5^1 \quad RHS = 1+4=5$$

S_n is true for $n=1$

step 2. Assume S_n is true for $n=k$.

$$\text{ie } S_k = 5^k \geq 1+4k, \quad k \in \mathbb{J}^+$$

step 3. Prove S_n is true for $n=k+1$
 $k \in \mathbb{J}^+$

$$RTP \quad 5^{k+1} \geq 1+4(k+1)$$

$$\text{ie } 5^{k+1} \geq 5+4k.$$

Consider $5^k \geq 1+4k$ (assumption)

$$5 \times 5^k \geq 5(1+4k) \quad \times \text{ by } 5.$$

$$5^{k+1} \geq 5+20k \geq 5+4K.$$

step 4. If $5^n \geq 1+4n$ for $n=k$.

then $5^n \geq 1+4n$ for $n=k+1$

since $5^n \geq 1+4n$ for $n=1$

then $5^n \geq 1+4n$ for $n=2, 3, 4, \dots$

thus $5^n \geq 1+4n$ for all $n \geq 1$.

(5)

4(b) Prove by induction $\arg z^n = n \arg z$
 $n \geq 1$
 $n \in \mathbb{J}^+$

Let S_n be the statement $\arg z^n = n \arg z$

step 1 for $n=1$

$$\arg z^1 = 1 \arg z.$$

S_n is true for $n=1$

step 2 Assume S_n is true for $n=k$.

$$\arg z^k = k \arg z.$$

step 3 Prove S_n is true for $n=k+1$

$$\arg z^k = k \arg z \quad \text{by assumption}$$

$$\arg z^{k+1} = \arg z^k \cdot z.$$

$$= \arg z^k + \arg z.$$

$$= k \arg z + \arg z.$$

$$= (k+1) \arg z$$

step 4 If $\arg z^n = n \arg z$ for $n=k$.

then " " " for $n=k+1$

since " " " for $n=1$

then " " " for $n=2, 3, 4, \dots$

thus $\arg z^n = n \arg z$ for all

$n \geq 1$

$n \in \mathbb{J}^+$

(4)